

II-Probabilistic Approaches Causality

Foundations of Neuro-Symbolic AI

Alex Goessmann

University of Applied Science Würzburg-Schweinfurt (THWS)

June 29, 2026

Outline

- ▶ Bayesian network hypergraphs

- ▶ Interventions

- ▶ Counterfactuals

Imagine a Bayesian network.

We can manipulate the hypergraph by changing the direction, and still find a Bayesian network representation of a distribution.

Causation is not (always) represented in the directionality!

The rungs of causation

Outline

► Bayesian network hypergraphs

► Interventions

► Counterfactuals

Outline

- ▶ Bayesian network hypergraphs
- ▶ Interventions
- ▶ Counterfactuals